

Project Review Meeting

on

Hotel Bar and Minibar Tracking

By

Sahasra Yeruva
Bhavika Pawar

2520080062
2520030616

Team Number: 1

Under the guidance of

D.Pavithra
Assistant Professor

Department of Computer Science and Engineering



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Outline of the Presentation

- Abstract
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- Problem Statement
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- Modules[Frontend web pages]
- Expected Outcomes of the Proposed Work
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Abstract

- The SmartSense Hotel Bar & Minibar Tracking System is a web based hospitality solution designed to automate and optimize minibar inventory management, beverage tracking, billing reconciliation, and room servicing operations.
- The system replaces traditional manual tracking methods with a centralized digital platform that provides:
 - 1.Real-time inventory monitoring
 - 2.Automated replenishment workflows
 - 3.Billing synchronization
 - 4.Activity tracking and auditing
 - 5.Low-stock notifications
 - 6.Analytical dashboards
- The solution improves operational efficiency, minimizes revenue leakage, reduces billing errors, and enhances guest satisfaction.

Introduction

- Hotels often face challenges in managing minibar inventory due to:
 1. Manual stock tracking
 2. Billing inaccuracies
 3. Delayed room servicing
 4. Inventory mismatches
 5. Lack of centralized monitoring
- SmartSense addresses these issues by providing a smart digital platform for tracking minibar consumption, managing inventory, generating reports, and coordinating staff operations in real time.

Problem Statement

Traditional hotel bar and minibar management relies on manual inventory tracking and disconnected communication between departments, leading to stock mismatches, billing errors, revenue leakage, delayed replenishment, and inefficient room servicing; therefore, there is a need for a centralized real time system that automates inventory monitoring, consumption tracking, billing reconciliation, and service coordination.

PROJECT FLOW DIAGRAM

SmartSense – Hotel Bar & Minibar Management System



OUTCOME: Improved inventory accuracy, timely replenishment, accurate billing, reduced wastage, and better guest experience.

Objectives with Justification

Objective 1:Digitize minibar inventory management.

Justification: Eliminates manual record maintenance.

Objective 2:Provide real-time inventory tracking.

Justification: Prevents stock shortages and delays.

Objective 3:Automate billing reconciliation.

Justification: Improves billing accuracy.

Objective 4:Enable replenishment workflow tracking.

Justification: Enhances staff coordination.

Objective 5:Provide analytics and reporting.

Justification: Supports data driven decision making.

Modules [Frontend web pages]

- Home Dashboard
- Minibar Log
- Consumption Form
- Replenishment Schedule
- Notifications
- Inventory Charts
- History Tracking

Expected Outcomes of the Proposed Work

- Improved minibar inventory accuracy
- Real-time stock monitoring
- Faster replenishment process
- Reduced inventory wastage
- Improved billing accuracy
- Better staff accountability
- Automated report generation
- Enhanced guest satisfaction
- Centralized hotel minibar management

Conclusion and Future Scope

- SmartSense provides a centralized platform for managing hotel minibar operations through automation, inventory tracking, billing integration, and analytics

Future Scope:

- AI-based demand forecasting
- IoT-enabled smart minibar sensors
- Mobile application integration
- Cloud deployment
- Predictive inventory analytics
- Multi-property hotel management support

References

- Weerasinghe, W.P.S.W., et al., "Research on Hotel Management System", Sri Lanka Institute of Information Technology, 2022.
- Hotel Inventory Management System, International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE).
- What an Important Room Mini Bar Management in Hotel PMS, MyHotelLine Blog – Inventory control, real-time monitoring, automated billing, and minibar analytics.
- Hotel Inventory Management Software: A Complete Guide, SiteMinder – Hotel inventory tracking, revenue optimization, reporting, and operational efficiency.
- Hotel Booking Web Application using MERN Stack, International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET), 2025

CO-WISE MAPPING TABLE

CO	CO DESCRIPTION (HANDOUT)	FEATURES IMPLEMENTED IN SMARTSENSE	
CO1	Foundations of Front-End Engineering & Framework Design (Imperative vs Declarative UI, Component-driven thinking, Rendering pipeline, Unidirectional data flow, Virtual DOM, Framework abstractions)		Declarative UI, Component-Based Architecture, Unidirectional Data Flow, Dynamic UI Updates, Reusable Front-End Components
CO2	JavaScript & TypeScript Engineering for Frameworks (ES6+ Features, Closures, Async/Await, Promise Chains, Modular Design, Functional Programming Concepts)		ES6+ Features, Arrow Functions, Modular Code Structure, Functional Programming, Array Methods (map/filter), Dynamic Data Processing
CO3	React Component Model (Component Development, Props, State, React Hooks, Controlled Components, Component Composition, Re-rendering Concepts)		React Components, Props Architecture, State Management (useState), React Hooks (useEffect), Controlled Forms, Component Reusability
CO4	State Architecture & API Integration (State Co-location, Derived State, Global State, Async Data Flow, Data Caching, API Service Layers)		Cart State Management, Derived State (Total Price, Cart Count), Data Synchronization, Billing Logic, Dynamic Order Summary
CO5	Routing, Forms, Accessibility & Performance (Routing, Protected Routes, Controlled Forms, Validation, Accessibility, Performance Optimization)		Multi-page Navigation, Billing Form, Checkout & Payment Options, Form Validation, Responsive Design, User-Friendly Interface
CO6	Build Systems, Testing & Deployment (Build Tools, Bundlers, Deployment, Testing, CI/CD, Performance Evaluation)		Scalable Front-End Architecture, Modular Folder Structure, Production-Ready UI, Deployment-Oriented Design, Maintainable Codebase

Thank you